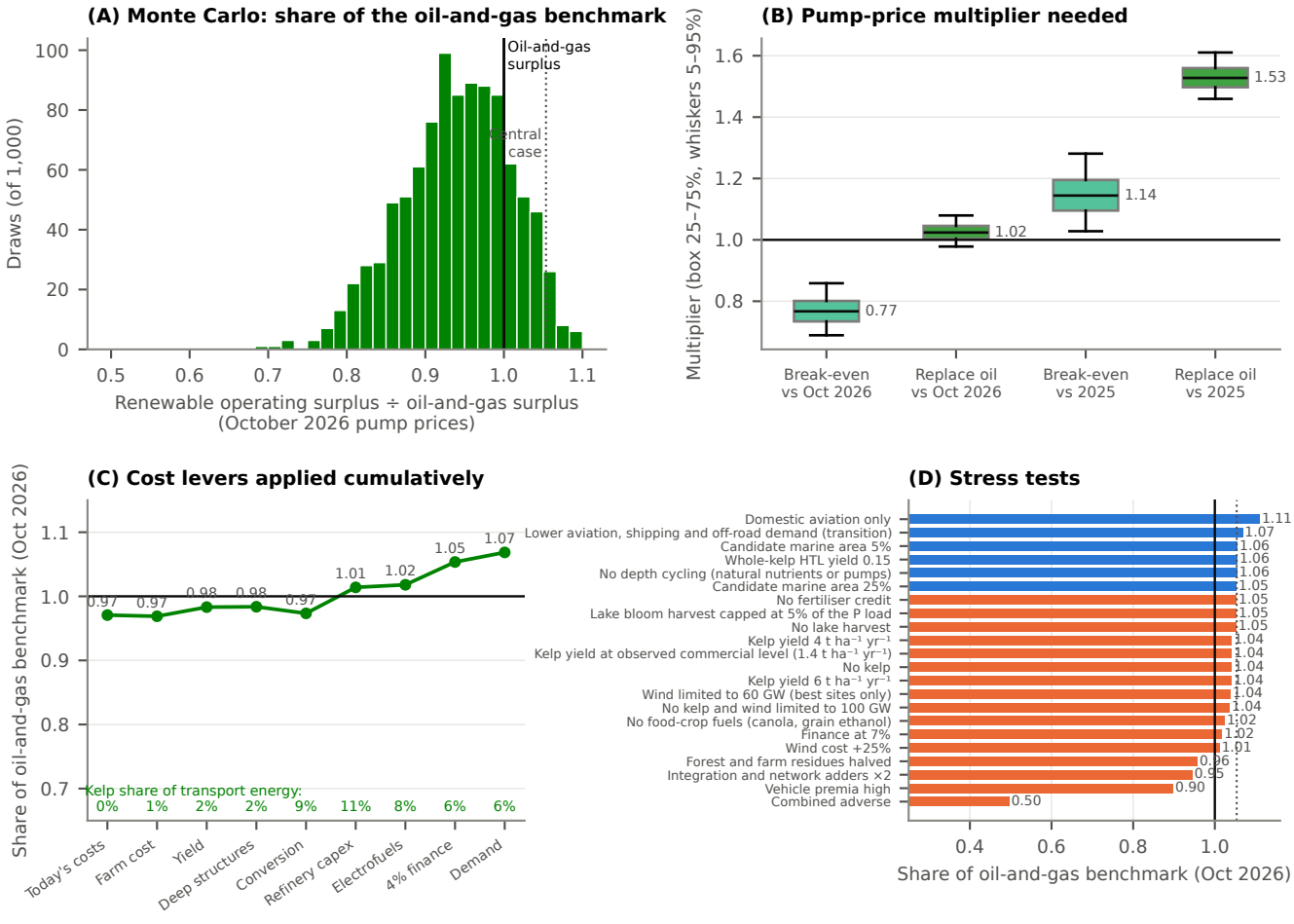
(A) Monte Carlo: share of the oil-and-gas benchmark
Draws (of 1,000)
Oil-and-gas surplus
Central case
Renewable operating surplus ÷ oil-and-gas surplus (October 2026 pump prices)
(B) Pump-price multiplier needed
Multiplier (box 25–75%, whiskers 5–95%)
0.77
1.02
1.14
1.53
Break-even vs Oct 2026
Replace oil vs Oct 2026
Break-even vs 2025
Replace oil vs 2025
(C) Cost levers applied cumulatively
Share of oil-and-gas benchmark (Oct 2026)
0.97
0.97
0.98
0.98
0.97
1.01
1.02
1.05
1.07
Kelp share of transport energy:
0% 1% 2% 2% 9% 11% 8% 6% 6%
Today's costs
Farm cost
Yield
Deep structures
Conversion
Refinery capex
Electrofuels
4% finance
Demand
(D) Stress tests
Domestic aviation only 1.11
Lower aviation, shipping and off-road demand (transition) 1.07
Candidate marine area 5% 1.06
Whole-kelp HTL yield 0.15 1.06
No depth cycling (natural nutrients or pumps) 1.06
Candidate marine area 25% 1.05
No fertiliser credit 1.05
Lake bloom harvest capped at 5% of the P load 1.05
No lake harvest 1.05
Kelp yield 4 t ha⁻¹ yr⁻¹ 1.04
Kelp yield at observed commercial level (1.4 t ha⁻¹ yr⁻¹) 1.04
No kelp 1.04
Kelp yield 6 t ha⁻¹ yr⁻¹ 1.04
Wind limited to 60 GW (best sites only) 1.04
No kelp and wind limited to 100 GW 1.04
No food-crop fuels (canola, grain ethanol) 1.02
Finance at 7% 1.02
Wind cost +25% 1.01
Forest and farm residues halved 0.96
Integration and network adders ×2 0.95
Vehicle premia high 0.90
Combined adverse 0.50
Share of oil-and-gas benchmark (Oct 2026)